

Active ants: dynamics

$$\begin{cases} \partial_t f = \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) = f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?

• Part 2 (Rakotomalala & W, in review, 2026):

◦ **Theorem.** There is a universal $c > 0$ such that if $\sigma_x < cPe/2\pi$ and σ_θ small enough, then there is a value of χ at which there exist two branches representing two families of stationary solutions: one constant in one spatial x -variable (lanes) and one swap-symmetric:
 $f(x_1, x_2, \theta) = f(-x_2, -x_2, -\theta - \pi/2)$ (spots)

◦ Existence of critical $\tau = \tau^*$ with super- to subcritical transition



For lanes and spots separately and $0 < \tau_s^* < \tau_l^*$

■ $\tau < \tau_s^*$ spots are locally nonlinearly stable (up to translation)

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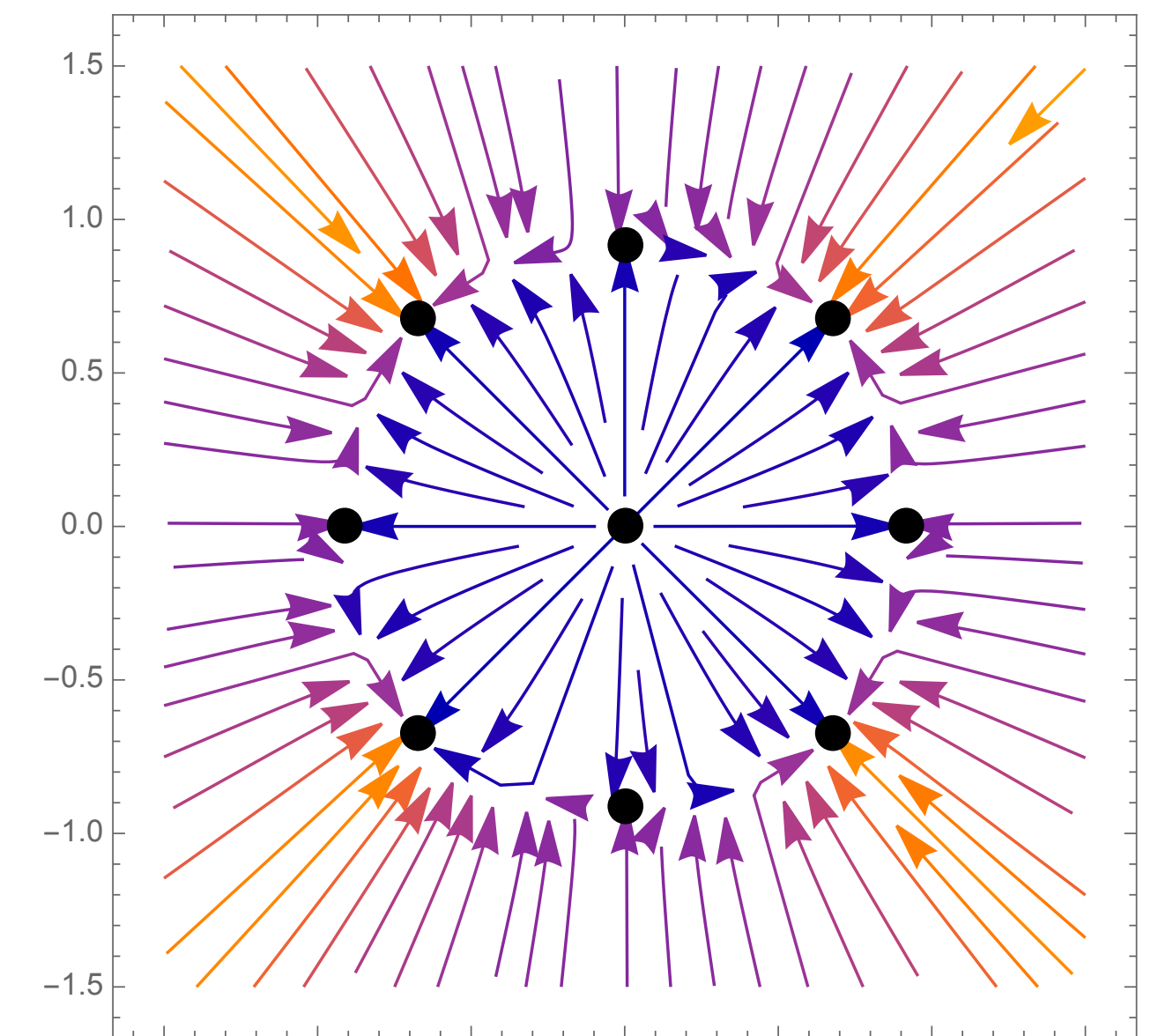
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- Part 2 (Rakotomalala & dW, in review, 2026):
 - **Theorem.** There is a universal $c > 0$ such that if $\sigma_x < c\text{Pe}/2\pi$ and σ_θ small enough, then there is a value of χ at which there exist two branches representing two families of stationary solutions: one constant in one spatial x -variable (lanes) and one swap-symmetric:
 $f(x_1, x_2, \theta) = f(-x_2, -x_1, -\theta - \pi/2)$ (spots)
 - Existence of critical $\tau = \tau^*$ with super- to subcritical transitions
 - For lanes and spots separately and $0 < \tau_s^* < \tau_l^*$
 - $\tau < \tau_s^*$ spots are locally nonlinearly stable (up to translation)

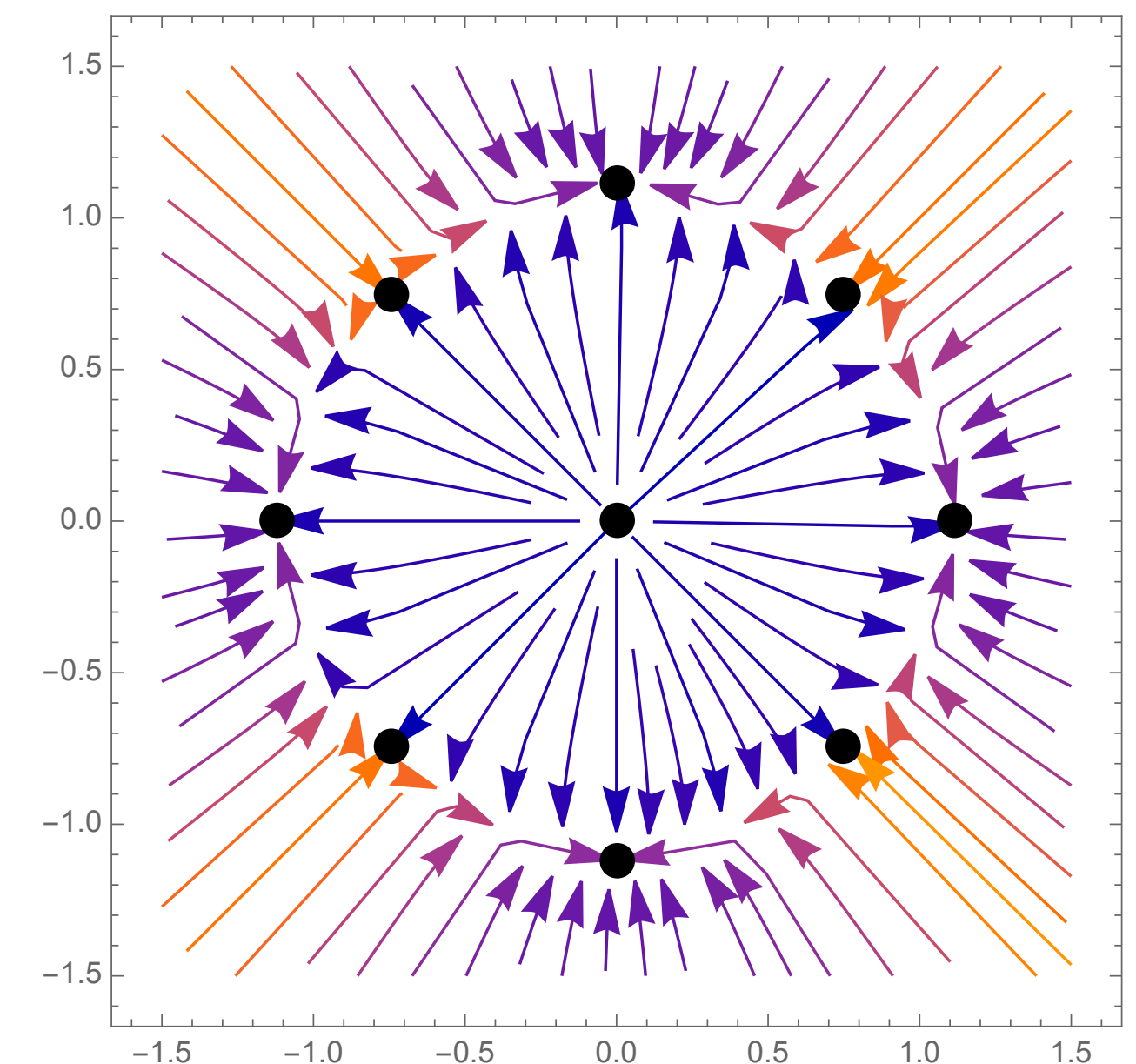
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- What happens as $t \rightarrow \infty$?
- Part 2 (Rakotomalala & dW, in review, 2026):
 - Existence of two families of non-trivial stationary states
 - Existence of critical $\tau = \tau^*$ super- to subcritical



$$\tau < \tau^*$$



$$\tau > \tau^*$$